GitHub Repository, Dataset and Demonstration Video

NeuroSonic project · Allwin Raj · Vellore Institute of Technology · 3 November 2026

> **Status.** The GitHub repository was created on 4 November 2026. It is **private** until the IPR cell has given advice on the planned invention disclosure (see the venue document); faculty members get access as collaborators. The demonstration video is stored in the same repository.

1. Links to Submit

Item	Link
GitHub repository (source code, tools, tests, results, documents)	https://github.com/allwinrajr2026-cell/NeuroSonic (private)
Dataset	Base-paper (WOAD) dataset: link to the authors' published Google Drive archive given in their repository; do not re-upload it. Project results: `outputs/` folder in the repository.
Demonstration video	https://github.com/allwinrajr2026-cell/NeuroSonic/blob/main/demo/NEUROSONIC_DEMO.mp4 (in the repository, `demo/NEUROSONIC_DEMO.mp4`)

2. Repository Contents

Content	Path in the repository
Application source	`neurosonic/`, `web/`, `neurosonic_app.py`
Base-paper re-implementation	`woad/`, `woad_detect.py`, `woad_replay.py`
Tools (benchmarks, evaluation, soak test, figures)	`tools/`
Tests	`tests/`
Result files	`outputs/benchmarks/`, `outputs/eval/`
Documentation and publication package	`docs/`, `PROJECT_DOCUMENTATION/`, `publication/`, `submission/`
Launch scripts and requirements	`START_NEUROSONIC.ps1`, `STOP_NEUROSONIC.ps1`, `requirements.txt`, `README.md`

Not uploaded	Reason
`data/tls/` (especially `neurosonic-ca.key`, `server.key`)	private keys of the local certificate authority: anyone with them could impersonate the server to phones that trust it

`data/neurosonic.sqlite3`, `logs/`	personal history and logs
Any API key (`GEMINI_API_KEY`)	secret; it is read from the environment and is not in the code
`.venv`, `.venv-gpu`, `__pycache__`, `backups/`	generated or private
WOAD dataset and original repository copy	third-party data and code: link to the authors' release instead
Model weights and voices (RF-DETR, SSDLite, Piper, Vosk, WOAD checkpoint)	large files with their own licences: document where to download them (the README already lists them)

The repository's `.gitignore` excludes: `data/`, `logs/`, `.venv*/`, `__pycache__/`, `backups/`, `*.pth`, `*.pt`, `*.onnx`, `models/`, `weights/`, `*.sqlite3`, `*.key`. The repository is licensed under AGPL-3.0 (the licence of the original WOAD code); the README lists the licences of all third-party components.

3. Demonstration Video: Shot Plan

Target length 8–10 minutes; screen recording of the laptop plus, where possible, a phone recording. Speak over the video, or add captions. Do not show the API key, the private key files, or personal history.

Time	Shot	What to say / show
0:00–0:45	Title slide (from `publication/NeuroSonic_Slides.pdf`)	problem, statistics, goal; "research prototype, not a safety device"
0:45–2:15	Base paper	WOAD architecture; run the WOAD replay on a recorded sequence; show the equivalence result and the evaluation table (0.218 / 0.173); explain why a phone camera cannot drive it
2:15–3:00	Launch	run `START_NEUROSONIC.ps1`; point out "RF-DETR … cuda (fp16), 200 ms" and the iPhone URL
3:00–5:30	Walking-path awareness	on the iPhone (or the laptop browser with the demo clip): Start; "appears clear"; a person enters from the side → "Caution. A person is moving into your path from the left."; pan over a static scene → no event; cover the lens → "cannot clearly assess"
5:30–7:00	Voice and navigation	push-to-talk "navigate to CMC Hospital"; confirm; offline route and spoken turn; disconnect the internet and repeat a function
7:00–8:00	Results	detector comparison, FP16, CPU-thread fix, soak test figures
8:00–9:00	Existing vs proposed	the A/B/C table from the demo document; pending iPhone and field tests

Only footage that is actually recorded may be presented as such. If the iPhone is not used, say that the browser emulation and the composited demo clip are shown.